Commentary

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From Deficits to Strengths: How Individual Differences in Neurodevelopment May Reflect Adaptations to Environmental Context

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Human brain development does not follow a predetermined, immutable path but instead reflects a dynamic interplay between flexible neurobiological programs and each individual's unique environmental context. As such, divergence from group-averaged developmental trajectories may be the norm rather than the exception, reflecting valuable adaptations to environmental circumstances rather than brokenness or failures. Adolescence represents one critical period of neurodevelopmental plasticity during which experience-dependent specialization and refinement of neural circuits support the successful transition to adulthood. During this transition, self-regulatory processes (e.g., affective and cognitive control) undergo protracted maturation, as adolescents learn to navigate novel and increasingly complex environments outside the immediate family. Though converging evidence underscores the profound influence of early-life environments in shaping both adaptive development of self-regulation and the risk for neuropsychiatric disorders characterized by difficulties with self-regulation, specific mechanisms linking environmental influences with cognitive and affective neurodevelopment remain obscure.

Investigating how environmental context shapes the timing and pace of specific neurodevelopmental processes may reveal how individual differences in cognitive and affective functioning emerge. To this end, in the current issue of *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, Tsomokos *et al.* (1) probe how socioeconomic disadvantage relates to internalizing problems via pubertal effects on cortical thinning in adolescence. Using a large, longitudinal sample from the Adolescent Brain Cognitive Development (ABCD) Study, Tsomokos *et al.* report sex-specific effects of socioeconomic status (SES) on pubertal timing and tempo. In females but not males, low SES was associated with earlier pubertal timing, which was related to faster cortical thinning, as well as slower pubertal tempo, which was related to slower cortical thinning. Puberty-related cortical thinning partially mediated the association between low SES and fewer internalizing problems in females, highlighting a novel pathway by which socioeconomic environments may be reflected in adolescent neurodevelopment and shape psychiatric risk and resilience.

Although the relationships among neurodevelopment, pubertal maturation, SES, and psychiatric risk are highly complex, recent work has begun disambiguating these associations. Previous work has described how puberty—comprising numerous metabolically expensive neuroendocrinological programs that are highly sensitive to environmental cues signaling resource availability—readies individuals for energy-intensive processes such as sexual development and potential reproductive or caregiving behaviors to follow (2). Tsomokos *et al.* (1) report that females growing up in socioeconomic environments with fewer resources available to meet their needs exhibit earlier cortical maturation, which could reflect an adaptive shift in developmental programs toward earlier reproductive, cognitive, and socioemotional readiness (3). The net effect of earlier pubertal timing and faster cortical thinning may represent a protective neurodevelopmental adaptation conferring resilience against internalizing problems; however, it will be important to test these effects in more representative samples. This interpretation is consistent with models from developmental and evolutionary biology positing that early-life stress may accelerate biological maturation and pressure an organism to strategically invest energy resources to support key developmental milestones.

While accelerated pubertal and cortical development may support the adaptive maturation of specialized skills, it is also worth noting that the stressors associated with socioeconomic disadvantage, and even the adaptive processes that may confer resilience, likely also carry downstream consequences that increase the risk of psychiatric symptoms. For example, goal-directed attention, group-level problem solving, and socioemotional attunement may need to mature rapidly to cope with stressful early-life environments, but the negative impacts of these stressors or the resource expenditure that enables accelerated maturation may predispose individuals to greater psychiatric risk. Universally labeling individual deviations from group-averaged neurodevelopmental trajectories as inherently maladaptive (a "deficit" model) may therefore obscure the varied and potentially adaptive developmental changes that facilitate successful navigation of difficult circumstances (a "strengths-based" perspective) (4). Identifying specific biological signatures underlying both adaptive shifts and negative consequences might facilitate the development of interventions targeting both individual-level treatments and broader societal policies.

Importantly, the findings by Tsomokos *et al.* (1) also clarify our understanding of the effects of socioeconomic resource availability by operationalizing SES according to household income-to-needs ratio. By considering both income and needs together, this measure more closely represents the availability of socioeconomic resources, providing enhanced specificity not captured by single measures, such as household income, that are agnostic to significant geographic

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variations in costs of living. This approach aligns well with growing interest in leveraging multidimensional measures to more comprehensively quantify each individual's complex environmental context, collectively referred to as an individual's exposome. Recognizing that environmental influences on cognitive and affective neurodevelopment are varied, additive, and interactive, the burgeoning field of exposomics seeks to comprehensively capture the rich diversity of each individual's physical/chemical exposures (e.g., lead poisoning, air pollution), psychosocial experiences (e.g., abuse, neglect), and external environments (e.g., SES, sociocultural factors) (5).

Future work may leverage exposomics to expand on the findings of Tsomokos *et al.* (1) and identify specific interactions across environmental domains; for example, socioeconomic resource availability may determine the neighborhood a child resides in, which in turn could affect the chemical toxins they are exposed to, the psychosocial stressors and sources of support they are likely to encounter, and their access to health care and education. In doing so, it will be important to acknowledge and critically evaluate the sociopolitical contexts that give rise to structural barriers—including limited access to high-quality schools, jobs, health care, food, public spaces, and stable housing—that can jeopardize healthy adolescent neurodevelopment. For example, current and historical systems of oppression, often rooted in racist and heteropatriarchal power structures, perpetuate acute and chronic physical, mental, and ecological harm against marginalized communities (6). Future studies investigating how environmental context becomes neurodevelopmentally embedded may therefore benefit from cross-disciplinary collaborations across the social sciences and public health as well as from direct community involvement. For example, community-based participatory research approaches may be used to directly involve community members with lived experiences not only as study participants but as collaborators across the lifespan of scientific investigations, from study design and measurement selection to interpretation and dissemination of results (7).

Future research building upon the findings of Tsomokos *et al.* (1) may also seek to integrate personalized neuroscience and precision psychiatry approaches to more closely examine how distinct environmental contexts shape person-specific brain development and confer psychiatric risk. An ongoing challenge in psychiatric medicine is that current diagnostic categories often include individuals with highly varied symptom profiles, and one-size-fits-all treatment approaches are often ineffective. To overcome this challenge, recent work in precision psychiatry has sought to identify novel biomarkers of transdiagnostic symptoms that can inform targeted clinical care. By leveraging comprehensive assessments of person-specific environmental context alongside cutting-edge neuroimaging techniques that can better capture individual differences in brain organization (e.g., precision functional mapping), future work may elucidate unique risk factors and biomarkers associated with psychiatric illness. Future work may also leverage multivariate assessments of puberty paired with dense longitudinal study designs to disambiguate the complex, nonlinear, and individual-specific trajectories of neuroendocrinological, social, affective, and cognitive maturation that contribute to the differences in cortical thinning reported by Tsomokos *et al.*, potentially revealing novel biomarkers and strategies for mitigating psychiatric risk.

Several notable psychological effects of puberty-related neuroendocrinological changes also warrant further exploration, including shifts in value attribution and attentional selection during adolescence as different features of the psychosocial environment become more salient. For example, adolescents become particularly sensitive to socioaffective signals (e.g., peer feedback) and adept at allocating attention toward these cues (social reorientation) (8). These shifts in prioritization during puberty may contribute to both adaptive cognitive and affective maturation as well as increased psychiatric risk, which is more strongly associated with puberty than with chronological age (9). Attentional selection also undergoes protracted maturation through adolescence, allowing the developing brain to facilitate processing of relevant information while learning to suppress irrelevant distractions. Selection of both external sensory information and internal cognitive and affective states may be strongly influenced by the characteristics of early-life environments. For example, a child may learn to shift attention rapidly toward potential threat signals or to maintain focus on reliable safety cues, and what a child attends to (e.g., sirens vs. lullabies) in turn may entrain neural tuning. This selective exposure—often co-regulated in early life by caregivers and later by peers (10)—may facilitate adaptive neurodevelopment to meet specific environmental demands, leading to interindividual heterogeneity in how attention selects certain thoughts, emotions, and behaviors over others. Longitudinal investigations of person-specific developmental trajectories for self-regulation, the broad skillsets of affective and cognitive control that are supported by value attribution and attentional selection, may therefore provide insight into the emergence of transdiagnostic psychiatric conditions characterized by self-regulation difficulties during adolescence.

Given sharp rises in socioeconomic disadvantage and the increased prevalence of psychiatric disorders among adolescents, it is imperative that we leverage recent empirical and theoretical advances in human neuroscience and biological psychiatry to better understand how environmental context shapes individualized neurodevelopmental trajectories. Together, Tsomokos *et al.* (1) and others have made important strides toward this goal by characterizing how early-life environments affect the timing and tempo of pubertal and cortical development to confer psychiatric risk. With these foundational studies, the field is well positioned to leverage individual differences in neurodevelopment as informative signals rather than extraneous noise or signs of brokenness to better inform public policy, neuroscientific investigations, and psychiatric care.

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References

1. Tsomokos DI, McLaughlin KA, Whittle S, Dhamala E, Mehta MA, Rakesh D (2026): Socioeconomic disadvantage, pubertal and brain development, and internalizing problems in adolescence: A longitudinal investigation. Biol Psychiatry Cogn Neurosci Neuroimaging 11:555–569.
2. Sisk CL, Foster DL (2004): The neural basis of puberty and adolescence. Nat Neurosci 7:1040–1047.
3. Ellis BJ, Sheridan MA, Belsky J, McLaughlin KA (2022): Why and how does early adversity influence development? Toward an integrated model of dimensions of environmental experience. Dev Psychopathol 34:1–25.
4. DeJoseph ML, Ellwood-Lowe ME, Miller-Cotto D, Silverman D, Shannon KA, Reyes G, *et al.* (2024): The promise and pitfalls of a strength-based approach to child poverty and neurocognitive development: Implications for policy. Dev Cogn Neurosci 66:101375.
5. Robinson H, Dave N, Barzilay R, Wagner A, Kells N, Keller AS (2025): The effect of the “exposome” on developmental brain health and cognitive outcomes. Neuropsychopharmacology 51:1–16.
6. Braveman P, Egerter S, Williams DR (2011): The social determinants of health: Coming of age. Annu Rev Public Health 32:381–398.
7. Collins SE, Clifasefi SL, Stanton J, The Leap Advisory Board, Straits KJE, Gil-Kashiwabara E, *et al.* (2018): Community-based participatory research (CBPR): Towards equitable involvement of community in psychology research. Am Psychol 73:884–898.
8. Crone EA, Dahl RE (2012): Understanding adolescence as a period of social–affective engagement and goal flexibility. Nat Rev Neurosci 13:636–650.
9. Pfeifer JH, Allen NB (2021): Puberty initiates cascading relationships between neurodevelopmental, social, and internalizing processes across adolescence. Biol Psychiatry 89:99–108.
10. Gee DG (2021): Early adversity and development: Parsing heterogeneity and identifying pathways of risk and resilience. Am J Psychiatry 178:998–1013.